

KOMSOMOLSK-ON-AMUR, Russian Federation

WMO: 315610

Lat: 50.533N

Lon: 137.033E

Elev: 22

StdP: 101.06

Time Zone: 10.00 (E10)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.8	(c) -34.6	(d) -39.8	(e) 0.1	(f) -36.8	(g) -37.2	(h) 0.1	(i) -34.4	(j) 8.9	(k) -15.1	(l) 7.9	(m) -15.6	(n) 0.5	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.5	(c) 28.7	(d) 22.1	(e) 27.3	(f) 21.3	(g) 25.9	(h) 20.3	(i) 23.3	(j) 27.0	(k) 22.4	(l) 25.8	(m) 21.5	(n) 24.7	(o) 3.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	16.8	25.7	21.2	15.9	24.4	20.3	15.0	23.6	69.8	27.1	66.2	25.7	63.0	24.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.9	6.1	-39.3	30.8	2.8	1.1	-41.3	31.6	-42.9	32.2	-44.4	32.8	-46.4	33.6
			-39.1	24.6	2.6	0.8	-41.0	25.1	-42.6	25.6	-44.0	26.0	-46.0	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-23.8	-19.9	-8.9	3.2	11.9	17.6	21.2	19.8	13.2	3.4	-9.5	-21.3
	DBStd	16.75	6.12	5.70	5.94	4.69	4.41	3.94	3.10	2.87	3.71	4.30	7.04	5.95
	HDD10.0	4497	1049	837	587	212	32	1	0	0	14	208	586	972
	HDD18.3	6657	1307	1071	845	455	203	60	12	17	158	463	836	1230
	CDD10.0	1088	0	0	0	7	90	228	346	304	109	4	0	0
	CDD18.3	206	0	0	0	0	3	38	100	63	2	0	0	0
	CDH23.3	1091	0	0	0	1	37	259	527	260	8	0	0	0
	CDH26.7	185	0	0	0	0	3	51	102	29	1	0	0	0
Wind	WSAvg	2.8	2.4	2.3	2.8	3.1	3.3	2.9	2.6	2.6	2.9	2.8	2.9	2.7
Precipitation	PrecAvg	795	32	23	38	48	79	82	114	121	83	70	56	48
	PrecMax	1015	86	88	85	107	180	127	202	218	155	169	127	121
	PrecMin	528	2	4	3	14	23	24	5	46	24	22	9	3
	PrecStd	121	21	17	22	22	34	25	45	47	32	38	33	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.5	-3.9	5.9	21.0	26.8	30.4	30.7	29.1	24.4	17.6	5.4	-3.9
		MCWB	-3.5	-5.6	1.5	11.4	17.3	22.8	24.0	22.4	19.1	12.2	2.7	-4.9
	2%	DB	-6.6	-6.6	4.0	16.7	24.0	28.1	29.1	27.4	21.9	14.4	3.4	-7.6
		MCWB	-7.6	-8.0	0.5	9.7	14.4	21.3	22.6	21.5	17.6	10.0	1.7	-8.6
	5%	DB	-11.5	-8.6	2.0	13.4	21.7	26.3	27.7	26.0	20.1	11.9	1.9	-10.0
		MCWB	-12.3	-9.7	-0.8	7.6	13.4	19.7	21.9	20.5	16.4	8.0	0.4	-10.7
	10%	DB	-14.3	-11.2	0.1	10.4	19.6	24.5	26.3	24.7	18.8	10.1	0.2	-12.1
		MCWB	-14.9	-12.1	-2.1	5.6	12.4	18.7	21.2	20.3	15.5	6.7	-1.1	-12.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.4	-5.6	2.4	12.7	17.6	23.5	24.9	23.7	19.9	12.3	3.3	-5.0
		MCDB	-2.5	-3.9	5.4	20.1	25.7	29.1	29.6	27.4	22.5	16.6	5.1	-4.0
	2%	WB	-7.8	-7.9	0.7	10.4	16.2	22.0	23.8	22.6	18.4	10.6	2.0	-8.4
		MCDB	-6.8	-6.6	3.3	15.3	22.2	27.2	27.4	26.0	21.1	13.6	3.1	-7.6
	5%	WB	-12.2	-9.7	-0.6	8.0	14.6	20.9	22.9	21.8	17.1	8.7	0.5	-10.6
		MCDB	-11.5	-8.7	1.8	13.8	19.4	25.4	26.0	24.7	19.5	11.6	1.7	-10.0
	10%	WB	-15.0	-12.0	-2.0	5.8	13.2	19.6	22.0	21.0	16.1	7.0	-1.0	-12.8
		MCDB	-14.4	-11.2	0.1	9.8	17.6	23.4	25.2	23.6	18.4	9.6	0.2	-12.1
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.9	11.2	9.2	10.2	9.3	7.5	7.5	8.5	8.4	7.5	8.8
		MCDBR	5.9	11.8	11.6	14.6	14.6	12.0	9.5	9.6	10.0	11.3	6.7	8.5
		MCWBR	5.5	10.5	8.5	7.8	7.0	5.8	4.2	4.5	5.6	7.1	5.2	8.1
	5% WB	MCDBR	5.9	11.4	10.0	14.2	12.2	10.4	8.0	7.9	9.0	10.1	5.7	8.5
		MCWBR	5.5	10.1	7.5	7.6	6.6	5.3	4.1	4.2	5.4	6.8	4.8	8.1
Clear-Sky Solar Irradiance	taub	0.280	0.319	0.394	0.511	0.577	0.508	0.506	0.433	0.394	0.416	0.355	0.285	
	taud	2.088	1.996	1.851	1.768	1.737	2.004	2.058	2.281	2.311	2.053	2.019	2.075	
	Ebn at Noon	722	782	777	726	699	761	755	799	791	660	599	643	
	Edh at Noon	81	117	164	201	217	168	157	120	104	112	88	71	
All-Sky Solar Radiation	RadAvg	1.59	2.81	4.35	5.01	5.12	5.59	5.14	4.44	3.48	2.22	1.39	1.21	
	RadStd	0.06	0.09	0.20	0.41	0.42	0.54	0.49	0.37	0.27	0.16	0.12	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page